Gokila Harini Karthikeyan

Machine Learning Engineer | AI Systems Researcher

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Education

California State University, Long Beach — Long Beach, CA
M.S. in Computer Science, GPA: 3.8/4.0 — *Expected May 2027*

Anna University — India
B.Tech. in Information Technology, GPA: 3.8/4.0 — *Sep 2020 – May 2024*

Experience

California State University, Long Beach — Long Beach, CA
Graduate Researcher — NSF CRII-Funded CARGen Project — *Jan 2026 – Jun 2026*

- Built an LLM test-generation system that combines static code analysis, API validation, compiler-guided repair, and automated execution; evaluated 8 LLMs on 6,300 methods from 4 Apache Java systems.
- Achieved 95.4% compilation, 81.5% test pass rate, and 68.0% method coverage with GPT-4o, improving pass rate by 20.0 percentage points over the strongest baseline.
- Designed a cost-aware router that learns from a 15% calibration sample which generation strategy works best for each model, reducing redundant pipeline cost by 38–44% across four LLMs.

TransUnion — Bengaluru, India
Associate SDET Engineer — *Jun 2024 – Jul 2025*

- Developed Groovy API automation across 10+ production Spring Boot modules and integrated regression suites into GitLab CI/CD, increasing coverage from 80% to 90% and reducing manual regression effort by 30–40%.
- Led JDK 21 migration validation and supported an Oracle-to-PostgreSQL migration, resolving compatibility and data-consistency issues while maintaining a 98% successful release-deployment rate and zero data loss.

TransUnion — Chennai, India
Application Development Intern — *Jan 2024 – Jun 2024*

- Built and tested a project-management platform with Java, Spring Boot, React, Spring Security, PostgreSQL, REST APIs, JUnit, and Mockito; reusable components reduced duplicate implementation effort by approximately 30%.

Selected AI Systems

TinyRobotLM — On-Device Language Model for Robot Control | GitHub — Oct 2026
PyTorch, NumPy, YOLO, Docker, NVIDIA Jetson Nano | Best Overall, MarinaHacks 6.0

- Trained a 3.2M-parameter transformer from scratch to translate natural-language commands into robot programs; created a leakage-free 41K-example dataset and achieved 93.8% exact match on unseen phrasings.
- Reimplemented inference in pure NumPy with KV caching as a framework-free 13 MB model, then ran it with YOLO on a 2019 Jetson Nano; increased perception frequency by over 25× while enforcing constrained decoding, grammar validation, and watchdog-compatible execution.

Audio Spectrogram Transformer Compression | GitHub — Mar 2026 – Jul 2026
PyTorch, structured pruning, knowledge distillation, GPU latency profiling

- Physically compressed an 86M-parameter transformer by 26.3%, cutting compute by 21.1% and T4 latency by 15.6% while retaining 0.9703 macro-F1 versus a 0.9706 baseline across three seeds.

Socrates' Duck — AI Socratic Debugging Assistant | GitHub — 2026
FastAPI, LangGraph, LiteLLM, TypeScript, React, SQLite

- Built a VS Code assistant that guides debugging through a deterministic eight-level hint policy, with provider-agnostic LLM orchestration, fail-closed answer-leakage detection, prompt-injection isolation, and local-only telemetry.

ASEP — Attention-Driven KV-Cache Eviction | GitHub — Mar 2026 – May 2026
LLM inference, GPU memory profiling, tiered-cache policy design

- Designed an attention-aware eviction policy after measuring a 7.6× transfer asymmetry on Tesla T4 hardware, reducing wasted evictions by 40% and promotion cost by 57% over an age-based policy.

Technical Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, C, Groovy **ML:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas, OpenCV
AI Systems: LLM evaluation, agents, LangGraph, LiteLLM, multi-model routing, guardrails, synthetic data, model compression
Engineering: FastAPI, Spring Boot, REST APIs, PostgreSQL, Docker, Kubernetes, AWS, Linux, GitLab CI/CD, GitHub Actions, Git
Recognition: Best Overall, MarinaHacks 6.0; Best Paper, ICNGCS 2023 (Springer CCIS); manuscripts submitted to ICSE 2027 and ICASSP 2027